Generated by Scopus AI, Sat Oct 10 2026

AI Traffic Control for Heavy Railways

RailSync: The AI Traffic Controller for Heavy Railways

Quick Reference

Key Findings Table

Theme	Key Insights	Supporting Citations
Enhanced Efficiency & Safety	AI models (XGBoost, deep learning, RL) optimize traffic, reduce delays, and improve safety via real-time anomaly detection and hazard prevention.	1 2 3
Integration & Digital Transformation	AI integration with Railway 4.0, IoT, digital twins, and edge computing enables real-time monitoring, dynamic scheduling, and proactive maintenance.	4 5 6
Lightweight AI Models	Lightweight and compressed AI architectures (e.g., MobileNet, YOLO variants) are essential for deployment in resource-constrained environments.	7 8 9
Explainability & Transparency	Explainable AI (XAI) frameworks (SHAP, LIME, fuzzy models) are critical for trust and regulatory compliance in safety-critical systems.	10 11 12
Research Gaps	Lack of holistic, integrated AI management platforms; need for domain-specific benchmarks and improved cross-modal data fusion.	1 13

Direct Answer

Current research on AI traffic control for heavy railways demonstrates significant advances in operational efficiency through predictive algorithms and dynamic scheduling, as well as effective safety enhancements enabled by real-time monitoring and collision prevention systems. However, challenges remain in achieving seamless integration with existing Railway 4.0 infrastructures, addressing computational constraints through lightweight models, and ensuring transparency in safety-critical systems with robust explainable AI frameworks. Future research should focus on creating holistic, fully integrated management platforms, developing standardized benchmarks, and enhancing multi-modal data fusion to further optimize railway traffic management.

Study Scope

- **Time Period:** 2018–2024
- **Disciplines:** Transportation engineering, computer science (AI/ML), systems engineering, safety-critical systems, industrial informatics
- **Methods:** Meta-analysis, empirical studies, simulation, case studies, algorithmic benchmarking, systems integration reviews

Assumptions & Limitations

- Most studies focus on specific subsystems (e.g., scheduling, maintenance) rather than holistic integration.
- Data quality and availability vary across regions and networks.
- Real-world deployment is often limited by legacy infrastructure and regulatory constraints.
- Explainability and transparency are still emerging areas, with limited standardized metrics.
- Computational benchmarks are context-dependent and may not generalize across all heavy rail environments.

Suggested Further Research

- Development of unified, explainable AI-driven traffic management platforms.
- Standardization of multi-modal data benchmarks for railway AI.
- Integration of large language models and generative AI for decision support and maintenance.
- Advancement of AIoT architectures and ultra-low latency (5G/6G) communication protocols.
- Domain-specific XAI frameworks and regulatory alignment for safety-critical applications.

1. Introduction

Background and Motivation

Heavy railway networks are facing unprecedented complexity due to increasing passenger and freight demands, urbanization, and the need for higher safety and reliability standards. Traditional traffic control systems, often reliant on rigid timetables and manual interventions, struggle to adapt to real-time disruptions, leading to inefficiencies, delays, and safety risks. The emergence of AI-based traffic control systems—leveraging machine learning, deep learning, and reinforcement learning—offers a transformative approach to address these challenges by enabling predictive analytics, dynamic scheduling, and real-time anomaly detection. These advancements are motivated by the need to maximize infrastructure capacity, minimize human error, and ensure robust, adaptive operations in increasingly dense and interconnected railway environments [1] [4] [13].

Scope of the Report

This report focuses on the state-of-the-art in AI-driven traffic control systems for heavy railways, examining the technologies, algorithms, and operational impacts that define the current landscape. It addresses the integration of AI with existing infrastructure, the challenges of deployment in safety-critical and resource-constrained environments, and the future directions necessary to achieve fully autonomous, explainable, and efficient railway traffic management [1] [14].

2. Theoretical Frameworks

State-of-the-Art AI Algorithms

AI technologies in heavy railway traffic control encompass a spectrum of approaches:

- **Machine Learning (ML):** XGBoost, random forests, and neural networks are widely used for predictive maintenance, fault detection, and delay prediction, achieving high accuracy (often >95%) in large-scale networks [2] [15] [16] [17].
- **Deep Learning (DL):** Convolutional and recurrent neural networks (e.g., LSTM, ConvLSTM) are applied for real-time monitoring, passenger flow prediction, and anomaly detection [18] [19] [20].
- **Reinforcement Learning (RL):** RL and multi-agent RL frameworks optimize dynamic scheduling, route planning, and adaptive control under uncertainty, particularly in high-density and disturbed traffic scenarios [5] [21] [22] [23].
- **Metaheuristic Algorithms:** Genetic algorithms, ant colony optimization, and hybrid bat algorithms are used for timetable rescheduling and delay minimization, outperforming traditional methods by up to 30% in delay cost reduction [24] [25] [26].

These algorithms collectively enable real-time adaptation, predictive analytics, and operational optimization, moving railway traffic management from static, timetable-based systems to dynamic, data-driven paradigms [1] [27] [28].

Predictive Maintenance and Fault Detection

AI models, particularly XGBoost and deep learning variants, have set new benchmarks in predictive maintenance:

- **Performance:** XGBoost achieves precision, recall, and F1-scores near 99.95% for component-level maintenance (e.g., air production units, bridges), enabling early anomaly detection and proactive interventions [15] [16] [17].
- **Explainability:** Techniques like SHAP and LIME are increasingly integrated to provide interpretable insights, balancing accuracy with transparency for stakeholder trust [29] [30].
- **Generalization:** While supervised learning dominates, hybrid and explainable models are needed to address complex, multi-modal datasets and ensure robustness across diverse railway subsystems [16] [29].

Dynamic Scheduling and Real-Time Traffic Management

AI-driven dynamic scheduling systems maintain safe intervals and optimize traffic flow by:

- **Rolling Horizon Optimization:** Continuously updating train orders, routes, and event times within a rolling planning window, enforcing safety constraints and enabling auto-rollback or human intervention as needed [21].
- **Real-Time Hazard Detection:** Integrating environmental and operational data for millisecond-level warnings and conflict resolution, especially under high-density conditions [5] [31].
- **Intelligent Decision Support:** Combining ML predictions with operations research optimization to improve throughput (by up to 32.5%) and reduce secondary delays (by 14.2%) [27].

Impact of Temporal Granularity on Prediction Accuracy

Temporal granularity is critical for short-term traffic flow prediction:

- **Multi-Resolution Features:** Incorporating multi-granularity temporal features (e.g., hourly, daily) improves model accuracy by at least 7.4% RMSE, with optimal granularity varying by station and scenario [19] [20] [32].
- **Trade-Offs:** Finer granularity enhances short-term accuracy but may introduce noise, while coarser granularity improves stability—requiring careful selection for each application [33] [34].

Synthesis: The theoretical landscape is characterized by a convergence of advanced AI algorithms, each addressing specific operational challenges in heavy railway traffic management. The integration of predictive, adaptive, and explainable models is central to achieving both efficiency and safety in complex, real-world environments.

3. Methods & Data Transparency

Data Sources and Model Validation

- **Empirical Data:** Studies utilize large-scale operational data from national railway networks (e.g., Indian Railways, Taiwan Railways), sensor networks, and simulation environments [2] [7] [35].
- **Benchmarking:** Performance is assessed using metrics such as accuracy, precision, recall, F1-score, RMSE, and throughput improvements, with cross-validation and real-world pilot deployments [15] [27] [36].
- **Simulation:** Metaheuristic and RL algorithms are often validated in simulated traffic scenarios to assess scalability, robustness, and real-time applicability [22] [28] [37].
- **Explainability:** XAI techniques are evaluated through user studies, interpretability metrics, and integration with digital twin simulations for validation in safety-critical contexts [10] [11] [12].

Data Quality and Connectivity

- **IoT and Edge Sensing:** Smart Railway IoT (SRIoT) and edge computing platforms ensure high-quality, real-time data acquisition and processing, with protocols for data integrity and cybersecurity [38] [39] [40].

- **Communication Protocols:** Advanced 5G/6G, SDN, and NFV architectures support ultra-low latency, reliable connectivity essential for AI deployment in dynamic railway environments [6] [41] [42].

Synthesis: The methodological rigor in AI-driven railway research is underpinned by diverse data sources, robust benchmarking, and a growing emphasis on explainability and real-world validation. However, challenges remain in standardizing datasets and ensuring interoperability across heterogeneous systems.

4. Critical Analysis of Findings

Implementation Challenges and Limitations

Computational Constraints and Lightweight AI Models

- **Necessity:** High computational demands of advanced AI models limit deployment on standard or low-power hardware, especially in large, resource-constrained networks [2] [35].

- **Solutions:** Lightweight CNNs, MobileNet-based architectures, and pruned YOLO variants achieve high accuracy (F2 > 0.89, recall > 0.9) with minimal computational overhead (<25 kB model size, <15 MFLOPs), enabling real-time deployment on edge devices [9] [35] [36].

- **Trade-Offs:** Simpler models (e.g., MLP) may lack discriminative power, necessitating a balance between model complexity and operational feasibility [43].

Explainability and Trust in Safety-Critical Applications

- **Importance:** The "black-box" nature of AI poses risks in safety-critical railway environments, making explainable AI (XAI) essential for transparency, trust, and regulatory compliance [10].

- **Techniques:** SHAP, LIME, fuzzy model-agnostic explanations, and visual methods (e.g., HeatMaps) are employed to clarify AI decisions and reduce operator cognitive load [11] [12].

- **Challenges:** Real-time constraints, heterogeneous data, and lack of standardized interpretability metrics complicate XAI implementation [12].

Data Quality, Connectivity, and Cybersecurity

- **Data Integrity:** IoT-edge computing, multidimensional data integrity checks, and advanced sensor networks ensure reliable data for AI models [38] [39].

- **Connectivity:** 5G/6G, SDN, and NFV architectures provide the necessary bandwidth and reliability for real-time AI operations [6] [41].

- **Cybersecurity:** Multi-layered security strategies (authentication, encryption, intrusion detection) are critical

to protect against evolving threats [44] [45].

Real-Time Infrastructure Monitoring under Power Limitations

- **Fiber Optic Sensing:** Distributed fiber optic sensing (DFOS) technologies, combined with lightweight AI models, enable real-time, long-range monitoring with minimal power consumption [46] [47].
- **Energy Optimization:** AI-driven power management and integration with renewable sources ensure stable operation in remote or power-limited settings [48] [49].

Synthesis: While AI offers transformative potential, practical deployment in heavy railway environments is constrained by computational, operational, and safety challenges. Lightweight models, explainable frameworks, robust data pipelines, and secure, low-latency communication are essential for scalable, trustworthy AI integration.

5. Real-World Implications

Integration with Existing Infrastructure and Safety Protocols

Integration with Railway 4.0 and Legacy Systems

- **Digital Transformation:** AI enables fully digital control of Railway 4.0 infrastructure, optimizing traffic management and enabling secure, autonomous operations via application interfaces [4] [50].
- **Legacy Integration:** Secure interfaces and digital twins facilitate the coexistence of AI-driven systems with legacy equipment, ensuring seamless transitions and compliance with safety standards [1] [5].

Role of Edge Computing and Network Virtualization

- **Edge Computing:** Offloads computationally intensive AI tasks to edge nodes, reducing latency and improving responsiveness in real-time traffic management [6] [51] [52].
- **SDN/NFV:** Enables flexible, scalable deployment of traffic management services, dynamic resource allocation, and secure, reliable communication [6] [53] [54].

AI, IoT, and Digital Twin Synergy

- **Real-Time Monitoring:** Integration of AI, IoT, and digital twins supports real-time simulation, predictive maintenance, and adaptive control, enhancing operational resilience and efficiency [55] [56] [57].
- **Risk Management:** Digital twins enable virtual testing and validation of AI decisions, improving early warning capabilities and reducing system delays [58] [59].

Safety Assurance: Automatic Train Protection and Obstacle Detection

- **ATP/ATO Integration:** AI-enabled train control systems incorporate automatic train protection (ATP) and

obstacle detection, using formal verification and real-time monitoring to ensure safety [60] [61] [62].

- **Multi-Sensor Fusion:** Combines GPS, IoT sensors, and computer vision for dynamic, multi-sensor safety management, reducing reliance on human operators and enabling proactive risk mitigation [62] [63].

Synthesis: The real-world impact of AI in heavy railways is most evident in its ability to integrate with and enhance existing infrastructure, support autonomous and safe operations, and enable predictive, adaptive management across complex, distributed systems.

6. Future Research Directions

Gaps in Integrated AI-Driven Traffic Management

- **Holistic Integration:** Current solutions are often siloed, addressing individual aspects (e.g., scheduling, maintenance) without holistic integration. There is a pressing need for unified platforms that enable cross-subsystem communication and coordinated decision-making [13] [14] [64].

- **Data Benchmarks:** Lack of standardized, multi-modal datasets hampers model validation and comparability across networks [13] [65].

Advancing Explainable AI for Safety and Transparency

- **Domain-Specific XAI:** Development of hybrid, interpretable architectures and domain-specific benchmarks is essential for regulatory compliance and operational trust in safety-critical environments [11] [12] [66].

- **Digital Twin Validation:** Integration of XAI with digital twins for adaptive training and validation in complex scenarios [12] [66].

AIoT Architectures and Ultra-Low Latency Communication

- **5G/6G Protocols:** Advanced communication protocols (e.g., blockchain-enhanced 5G-R, quantum/lattice cryptography) are proposed for secure, ultra-low latency operations [41] [42] [45] [67].

- **Edge-Cloud Collaboration:** Hybrid architectures combining edge and cloud resources for scalable, real-time AI deployment [68].

Application of Large Language Models and Generative AI

- **Decision Support:** LLMs and generative AI can automate information processing, maintenance documentation, and real-time decision support, though their application in railways is still nascent [69] [70] [71] [72].

- **Human-Centric Design:** Integration with VR/MR and human-in-the-loop systems to enhance usability and effectiveness [73] [74].

Towards Fully Integrated AI-Driven Railway Platforms

- **Multi-Agent Systems:** Use of multi-agent and digital twin frameworks for dynamic, adaptive traffic management [75] [76].
- **Closed-Loop Control:** Real-time, closed-loop integration of IoT sensing, AI analytics, and operational control for unified management [77].
- **Scalability:** Lightweight, computationally efficient frameworks for large, resource-constrained networks [2].

Synthesis: The future of AI in heavy railway traffic management lies in the convergence of explainable, integrated, and scalable platforms, supported by advanced communication protocols and human-centric design. Addressing current research gaps will require interdisciplinary collaboration, standardization, and continuous innovation.

Conclusion

AI-driven traffic control systems are reshaping the landscape of heavy railway management, offering unprecedented gains in efficiency, safety, and adaptability. The integration of advanced algorithms, lightweight models, explainable frameworks, and robust communication infrastructures is enabling a transition from static, manual operations to dynamic, autonomous, and resilient systems. However, realizing the full potential of AI in this domain requires overcoming significant challenges in holistic integration, data standardization, explainability, and real-world deployment. Future research must prioritize the development of unified, explainable, and scalable platforms, leveraging the synergies of AI, IoT, digital twins, and next-generation communication technologies to achieve truly autonomous and safe railway operations.

Suggested Further Research

- **Unified AI Platforms:** Develop and validate holistic, explainable AI-driven traffic management systems that integrate scheduling, maintenance, safety, and risk management.
- **Standardized Benchmarks:** Create and share multi-modal, domain-specific datasets for benchmarking AI models in railway contexts.
- **LLM and Generative AI Integration:** Explore the application of large language models and generative AI for decision support, maintenance, and information management.
- **AIoT and 5G/6G Architectures:** Advance research on AIoT frameworks and ultra-low latency communication protocols for real-time, secure railway operations.
- **Domain-Specific XAI:** Design and implement explainable AI frameworks tailored to the unique requirements of safety-critical railway environments.

Note: All claims and findings are supported by the referenced empirical studies and meta-analytical evidence as indicated by the inline citations.

References

1. Prospects of Applying Artificial Intelligence in Microprocessor Systems for Railway Traffic Control Kulbovskyi, I., Tkachuk, M. Transport Means - Proceedings of the International Conference, 2024 https://www.scopus.com/pages/publications/85216762691?origin=scopusAI

2. Artificial Intelligence for Indian Railways Operation Optimization and Predictive Maintenance Dabhade, D.N., Sham Badhe, B., Balagide, S.S., (...), Wagh, K. 2026 2nd International Conference on Computing, Communication and Green Engineering, CCGE 2026, 2026 https://www.scopus.com/pages/publications/105044710154?origin=scopusAI

3. AI Based Rush Collision Prevention in Railways Viji, K., Chitra, K., Aravind, A.M., (...), Tarika, B. 2025 IEEE International Conference on Advanced Computing Technologies, ICACT 2025, 2025 https://www.scopus.com/pages/publications/105033658157?origin=scopusAI

4. ARTIFICIAL INTELLIGENCE IN OPERATIONAL APPLICATIONS OF RAILWAY INFRASTRUCTURE MANAGER Šrámek, P., Široký, J., Vojtek, M. Acta Polytechnica CTU Proceedings, 2023 https://www.scopus.com/pages/publications/85179607483?origin=scopusAI

5. Application of Artificial Intelligence Technology in Railway Signal Automatic Control Nie, J., Shi, Q., Zhang, M. Proceedings - 2026 11th Asia Conference on Power and Electrical Engineering, ACPEE 2026, 2026 https://www.scopus.com/pages/publications/105041616720?origin=scopusAI

6. Enabling Network Technologies For Flexible Railway Connectivity Narouwa, M., Mendiboure, L., Badis, H., (...), Langar, R. IEEE Access, 2024 https://www.scopus.com/pages/publications/85207708711?origin=scopusAI

7. LDRSM-A: A Lightweight Audio-Based Diagnosis Network for Railway Point Machines Liu, Z. 2025 7th International Conference on System Reliability and Safety Engineering, SRSE 2025, 2025 https://www.scopus.com/pages/publications/105032878095?origin=scopusAI

8. An Efficient and Scalable Real-Time Vehicle Surveillance System for Intelligent Traffic Management on Resource-Constrained Devices Vishwa, R., Sachin Arul, A., Subha, J. 2026 International Conference on Smart Futuristic Technology, ICSFT 2026, 2026 https://www.scopus.com/pages/publications/105041613762?origin=scopusAI

9. Fasteners quantitative detection and lightweight deployment based on improved YOLOv8 Bai, T., Duan, J., Wang, Y., (...), Zong, H. Review of Scientific Instruments, 2024 https://www.scopus.com/pages/publications/85205798600?origin=scopusAI

10. A Systematic Review on Explainable Artificial Intelligence in Railway: Taxonomy, Application, and Prospect Zhao, Z., Xun, J., Cao, Y., (...), Cai, Y. IEEE Transactions on Intelligent Transportation Systems, 2026 https://www.scopus.com/pages/publications/105037749196?origin=scopusAI

11. An Explainable Control Framework (XCF): A Novel Fuzzy Model-Agnostic Explanation (FMAE) Approach with Case Study Yin, F., Watson, D., Lam, H.-K. 2025 10th IEEE International Conference on Advanced Robotics and Mechatronics, ICARM 2025, 2025 https://www.scopus.com/pages/publications/105031783936?origin=scopusAI

12. Shedding Light on Explainable AI: Insights, Challenges, and the Future of Infrastructure Management Hu,

Y., Atta, Z., Rahman, T.U., (...), Zaheer, Q. ISPRS International Journal of Geo-Information, 2026 https://www.scopus.com/pages/publications/105034537579?origin=scopusAI

13. Artificial Intelligence Applied on Traffic Planning and Management for Rail Transport: A Review and Perspective Zhang, J., Zhang, J. Discrete Dynamics in Nature and Society, 2023 https://www.scopus.com/pages/publications/85167732604?origin=scopusAI

14. A literature review of Artificial Intelligence applications in railway systems Tang, R., De Donato, L., Bešinović, N., (...), Wang, Z. Transportation Research Part C: Emerging Technologies, 2022 https://www.scopus.com/pages/publications/85129622349?origin=scopusAI

15. Proactive Failure Prediction in Train Air Production Units Using XGBoost for Enhanced Safety Yahiaoui, L., Aggoune, A., Amrani, K., (...), Hamouid, K. 21st International Wireless Communications and Mobile Computing Conference, IWCMC 2025, 2025 https://www.scopus.com/pages/publications/105011363752?origin=scopusAI

16. Automated prognostics and diagnostics of highspeed railway faults using machine learning approaches Kongsap, P., Kaewunruen, S. Transportation Engineering, 2026 https://www.scopus.com/pages/publications/105037601146?origin=scopusAI

17. A Machine Learning-Based Algorithm for the Prediction of Eigenfrequencies of Railway Bridges Grunert, G., Grunert, D., Behnke, R., (...), Challagonda, S.R. International Journal of Structural Stability and Dynamics, 2025 https://www.scopus.com/pages/publications/85206946720?origin=scopusAI

18. Exploration of Machine Learning-Enabled Prediction and Control Algorithms for Railway Traffic Management Zheng, Y. Lecture Notes in Networks and Systems, 2024 https://www.scopus.com/pages/publications/85213353446?origin=scopusAI

19. A Short-term OD Passenger Flow Prediction Method for Rail Transit Based on Multi-resolution and Multi-granularity Spatio-temporal Feature Extraction Yang, Y., Pan, B., Liu, J., (...), Zhang, A. Tiedao Xuebao/Journal of the China Railway Society, 2024 https://www.scopus.com/pages/publications/105009578233?origin=scopusAI

20. MSTFormer: Multi-Scale Spatio-Temporal Transformer with Hierarchical Granularity Fusion for Traffic Flow Prediction Zhang, J., Quan, W., Liu, C., (...), Wang, H. Proceedings of 2026 6th International Conference on Applied Mathematics, Modelling and Intelligent Computing, CAMMIC 2026, 2026 https://www.scopus.com/pages/publications/105049702110?origin=scopusAI

21. Application and Implementation of Adaptive Railway Signaling Technology Yi, C. Proceedings of 2026 International Symposium on Robotics, Automation, and Mechatronics, RAM 2026, 2026 https://www.scopus.com/pages/publications/105042433808?origin=scopusAI

22. A Deep Reinforcement Learning Approach for Optimal Scheduling of Heavy-haul Railway Wu, T., Dong, W., Ye, H., (...), Ji, Y. IFAC-PapersOnLine, 2023 https://www.scopus.com/pages/publications/85184959218?origin=scopusAI

23. Real-Time Traffic Optimization: A Graph-based Reinforcement Learning Framework Sumathi, N., Navamani, G. Proceedings of 8th International Conference on Inventive Computation Technologies, ICICT 2025, 2025 https://www.scopus.com/pages/publications/105007428285?origin=scopusAI

24. A comparison of algorithms for minimising delay costs in disturbed railway traffic scenarios Fan, B., Roberts, C., Weston, P. Journal of Rail Transport Planning and Management, 2012 https://www.scopus.com/pages/publications/84870060742?origin=scopusAI

25. Optimal Scheduling for Delay Management in Railway Network Using Hybrid Bat Algorithm Dalapati, P., Paul, K. Lecture Notes in Electrical Engineering, 2021 https://www.scopus.com/pages/publications/85101531171?origin=scopusAI

26. A Real-Time Railway Traffic Management Approach Preserving Passenger Connections Sharma, B., Pascariu, B., Pellegrini, P., (...), Chaudhary, N. IEEE Access, 2024 https://www.scopus.com/pages/publications/85195388830?origin=scopusAI

27. AI-Driven Railway Section Management: An Intelligent Decision Support System using Neural Networks, Random Forest and an Operations Research Interface Perumal, I., Kalaivani, P., Srivarsne, H., (...), Sowmiya Shree, M. 7th International Conference on Electronics and Sustainable Communication Systems, ICESC 2026 - Proceedings, 2026 https://www.scopus.com/pages/publications/105052189955?origin=scopusAI

28. A variable neighbourhood search for fast train scheduling and routing during disturbed railway traffic situations Samà, M., D'Ariano, A., Corman, F., Pacciarelli, D. Computers and Operations Research, 2017 https://www.scopus.com/pages/publications/84959308471?origin=scopusAI

29. Trade-Off Between Interpretability and Accuracy: How Can XAI Build Trust in Track Geometry Predictive Maintenance? Mansouri, S.A., Dziedzic, R., Licciardello, R., (...), Nitti, M. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105023144406?origin=scopusAI

30. Scenario Identification for Autonomous Trains: An Explainable Anomaly Detection Framework with XGBoost and SHAP Zhai, W., Yan, F. 38th Chinese Control and Decision Conference, CCDC 2026, 2026 https://www.scopus.com/pages/publications/105043923442?origin=scopusAI

31. Balancing a one-way corridor capacity and safety-oriented reliability: A stochastic optimization approach for metro train timetabling Yin, J., Yang, L., Zhou, X., (...), Gao, Z. Naval Research Logistics, 2019 https://www.scopus.com/pages/publications/85065203281?origin=scopusAI

32. Analysis of prediction accuracy under the selection of optimum time granularity in different metro stations Li, P., Ma, C., Ning, J., (...), Zhu, C. Sustainability (Switzerland), 2019 https://www.scopus.com/pages/publications/85073628464?origin=scopusAI

33. Analyzing the Impact of Temporal Measurement Granularity on Network Traffic Forecasting De Castro, I.S.F., Ferreira, M.C.M.M., Filho, G.P.R., (...), Gomes, R.L. IEEE International Conference on High Performance Switching and Routing, HPSR, 2026 https://www.scopus.com/pages/publications/105046722294?origin=scopusAI

34. IoT Data-Driven Energy Prediction for Smart Buildings: A Comparative Analysis of Machine Learning Models Aytac, M., Shayea, I., Aldasheva, L., Amirova, A. SIST 2026 - 2026 IEEE 6th International Conference on Smart Information Systems and Technologies, Conference Proceedings, 2026 https://www.scopus.com/pages/publications/105045608423?origin=scopusAI

35. Intelligent Railway Block System Design with Lightweight CNN and RFID Chang, Y.-Y., Hung, K.-J., Perng, J.-W. 2024 International Automatic Control Conference, CACS 2024, 2024 https://www.scopus.com/pages/publications/85214989160?origin=scopusAI

36. YOLOv8-ECA-SPD-Lite for Cantilever Bolt Defect Detection in Railway Catenary Systems Wang, Y., Chen, M., Guo, L., (...), Wu, M. IEEE Access, 2025 https://www.scopus.com/pages/publications/105015214494?origin=scopusAI

37. Reinforcement learning in railway research and applications: A review Ni, H., Fu, Q., Easton, J., (...), Arvanitis, T.N. Transportation Research Interdisciplinary Perspectives, 2026 https://www.scopus.com/pages/

publications/105051367590?origin=scopusAI

38. Review of research on smart railway Internet of things Xie, J., Zhang, Z., Lin, W., (...), Qu, Y. Chinese Journal on Internet of Things, 2025 https://www.scopus.com/pages/publications/105019613110?origin=scopusAI

39. Multidimensional Data Integrity Checking Scheme for IoT-Edge Computing-Assisted Intelligent Railway Systems Yan, Z., Zhang, W., Wang, X., Khan, M.K. IEEE Transactions on Vehicular Technology, 2025 https://www.scopus.com/pages/publications/85214834288?origin=scopusAI

40. Multi-antenna transmission schemes in high speed railway communication Li, Z., Wang, F., Zhong, Z. IET Conference Publications, 2013 https://www.scopus.com/pages/publications/84902211423?origin=scopusAI

41. Research on High-speed Railway RSSP Secure Communication Protocol Enhanced by Blockchain Consensus Chen, Y., Zhang, B., Xin, Z. Tiedao Xuebao/Journal of the China Railway Society, 2024 https://www.scopus.com/pages/publications/105004256843?origin=scopusAI

42. Period-Aware Scheduling in Time-Sensitive Networking for Cloud-Based Computer Interlocking Lan, M., Chen, J., Liu, J., Xu, Z. IEEE Internet of Things Journal, 2026 https://www.scopus.com/pages/publications/105048442988?origin=scopusAI

43. A Study on AI-Based Monitoring Methods for Railway Signaling Impedance Bonds Jung, I.-B., Choi, S.H. Transactions of the Korean Institute of Electrical Engineers, 2025 https://www.scopus.com/pages/publications/105029898573?origin=scopusAI

44. Cybersecurity in intelligent railway systems: Taxonomy, research trends, challenges, and future directions Abukeshek, M., Al-Mhiqani, M., Parkinson, S., (...), Bearfield, G. Computers and Electrical Engineering, 2026 https://www.scopus.com/pages/publications/105028933913?origin=scopusAI

45. Blockchain based secure 6G network slicing identity verification and authorization Lin, T. Kuwait Journal of Science, 2026 https://www.scopus.com/pages/publications/105041203268?origin=scopusAI

46. Fiber-Optical-Sensor-Based Technologies for Future Smart-Road-Based Transportation Infrastructure Applications Senkans, U., Silkans, N., Merijs-Meri, R., (...), Bobrovs, V. Photonics, 2026 https://www.scopus.com/pages/publications/105031530261?origin=scopusAI

47. A collection of machine learning assisted distributed fiber optic sensors for infrastructure monitoring Karapanagiotis, C., Hicke, K., Krebber, K. Technisches Messen, 2023 https://www.scopus.com/pages/publications/85147204328?origin=scopusAI

48. Intelligent Monitoring and Power Management of Optical Telecommunication Networks Using Artificial Intelligence Yurievna, A.N. Proceedings of the 2025 Optical Communication, Photonics, Telecommunications, and Intelligent Machine Applications, OPTIMA 2025, 2025 https://www.scopus.com/pages/publications/105035396358?origin=scopusAI

49. AI-Driven Energy Optimization: Integrating Smart Meters, Controllers, and Cloud Analytics for Efficient Urban Infrastructure Management Whig, P., Balantrapu, S.S., Whig, A., (...), Dutta, P.K. IET Conference Proceedings, 2024 https://www.scopus.com/pages/publications/105003585144?origin=scopusAI

50. RESEARCH ON TECHNICAL AND LEGISLATIVE ASPECTS OF THE “INTELLIGENT END OF TRAIN” SYSTEM Leso, M., Musil, T., Zoubek, O., (...), Tomášek, M. Acta Polytechnica CTU Proceedings, 2026 https://www.scopus.com/pages/publications/105048716829?origin=scopusAI

51. Edge Computing for AI-Optimized Traffic Management in Autonomous Cars Ashirova, A., Shojonov, M., Meylibayevich, S.K., Patro, P. 2025 2nd International Conference on New Frontiers in Communication, Automation, Management and Security, ICCAMS 2025, 2025 https://www.scopus.com/pages/publications/105030121669?origin=scopusAI

52. 5G-enabled Edge Intelligence for Autonomous Train Control: A Practical Perspective Chen, C., Zhu, L., Zhao, H. IEEE Conference on Intelligent Transportation Systems, Proceedings, ITSC, 2022 https://www.scopus.com/pages/publications/85141874256?origin=scopusAI

53. A MAPE-K and Queueing Theory Approach for VNF Auto-scaling in Edge Computing Silva, T.P., Batista, T.V., Battisti, A.L., (...), Pires, P.F. 2023 IEEE 12th International Conference on Cloud Networking, CloudNet 2023, 2023 https://www.scopus.com/pages/publications/85191255915?origin=scopusAI

54. Towards Intelligent, Programmable, and Open Railway Networks Pencheva, E., Atanasov, I., Trifonov, V. Applied Sciences (Switzerland), 2022 https://www.scopus.com/pages/publications/85129146300?origin=scopusAI

55. Recent Advances in Artificial Intelligence for Improving Railway Operations Vatakov, V., Pencheva, E., Dimitrova, E. 2022 30th National Conference with International Participation, TELECOM 2022 - Proceedings, 2022 https://www.scopus.com/pages/publications/85147542419?origin=scopusAI

56. TRANSFORMING RAILWAY TRANSPORTATION: THE ROLE OF EMERGING TECHNOLOGIES IN EFFICIENCY, SAFETY, AND SUSTAINABILITY Zaheer, Q., Qiu, S., Atta, Z., (...), Wang, J. Journal of Civil Engineering and Management, 2026 https://www.scopus.com/pages/publications/105036592452?origin=scopusAI

57. Digital Twins and Railway Networks Maintenance: A Partnership for Efficiency and Safety Mhalla, A., Collart-Dutilleul, S. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105037196004?origin=scopusAI

58. Railway Geological Hazard Simulation Based on Digital Twin Technology and AI Models Wang, Y., Yang, X., Tang, X. Proceedings of SPIE - The International Society for Optical Engineering, 2026 https://www.scopus.com/pages/publications/105040206131?origin=scopusAI

59. Revolutionizing railway systems: A systematic review of digital twin technologies Thompson, E.A., Lu, P., Alimo, P.K., (...), Abbew, C.K. High-speed Railway, 2025 https://www.scopus.com/pages/publications/105007556529?origin=scopusAI

60. A Formal Model of Train Control with AI-Based Obstacle Detection Gruteser, J., Geleßus, D., Leuschel, M., (...), Vu, F. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2023 https://www.scopus.com/pages/publications/85174444013?origin=scopusAI

61. Fail-Safe On-Board Communication for Automatic Train Protection Eschermann, B., Kirrmann, H. Informatik aktuell, 1994 https://www.scopus.com/pages/publications/105046974029?origin=scopusAI

62. A railway collision avoidance system exploiting ad-hoc inter-vehicle communications and GALILEO Strang, T., Zu Hörste, M.M., Gu, X. 13th World Congress on Intelligent Transport Systems and Services, 2006 https://www.scopus.com/pages/publications/84950169313?origin=scopusAI

63. Kavach: A Technological and Strategic Framework for Railway Safety in India on Automatic Train Protection (ATP) Practices Muralidhar, P., Jaisai, T. Lecture Notes in Civil Engineering, 2025 https://www.scopus.com/pages/publications/105029897361?origin=scopusAI

64. Track Wise: Smart Railway Management Jitesh, K., Lakhina, G., Kolhar, L.S., (...), Taneja, K. International Conference on AI-Driven Innovations and Applications, IC-AIDA 2026 - Conference Proceedings, 2026 https://www.scopus.com/pages/publications/105044308374?origin=scopusAI

65. A review of applications of AI in monitoring, inspection, and maintenance of railway tracks Khajehdezfuly, A., Azizipour, H., Kaewunruen, S. Journal of Industrial Information Integration, 2025 https://www.scopus.com/pages/publications/105020966881?origin=scopusAI

66. Exploring the role of explainable AI in urban rail transit operations and passenger services Li, J. Proceedings of SPIE - The International Society for Optical Engineering, 2026 https://www.scopus.com/pages/publications/105033072403?origin=scopusAI

67. Function Management in Future Railway Mobile Communication Systems Atanasov, I., Trifonov, V., Pencheva, E. Procedia Computer Science, 2022 https://www.scopus.com/pages/publications/85163618052?origin=scopusAI

68. LoRa Mesh-5G Integrated Network for Trackside Smart Weather Monitoring Chen, Y., Shi, G., Al-Quraan, M., (...), Imran, M. IEEE Transactions on Vehicular Technology, 2024 https://www.scopus.com/pages/publications/85184337678?origin=scopusAI

69. How Large Language Models Can Be Utilized in the Railway Maintenance Processes Madhushika, C., Karim, R., Söderholm, P., (...), Kour, R. Lecture Notes in Mechanical Engineering, 2026 https://www.scopus.com/pages/publications/105041725107?origin=scopusAI

70. Leveraging AI-Powered Large Language Models to Improve Operational Safety and Efficiency in the Metal and Steel Industry Zhang, Y., Mohammed, Z., Marfatia, Z., Shah, A. AISTech - Iron and Steel Technology Conference Proceedings, 2025 https://www.scopus.com/pages/publications/105008452715?origin=scopusAI

71. Building Artificial Intelligence Agents with Large Language Models for Autonomous Interpretation of Equipment Sensor Data: P2 Carvalho, A., Boyle, F., Daly, J., Walsh, J. Proceedings of the 2025 IEEE International Conference on Computation, Big-Data and Engineering, ICCBE 2025, 2025 https://www.scopus.com/pages/publications/105031092016?origin=scopusAI

72. Predictive maintenance report generation using large language models Kernács, M., Hornyák, O. Pollack Periodica, 2026 https://www.scopus.com/pages/publications/105042091465?origin=scopusAI

73. A Human-Centric Model for Sustainable Asset Management in Railway: A Case Study Kour, R., Castaño, M., Karim, R., (...), Granström, R. Sustainability (Switzerland), 2022 https://www.scopus.com/pages/publications/85122909534?origin=scopusAI

74. A Generative AI Copilot for Supporting Plant Maintenance Operations in a Steel Plant Cestari, L., Felline, G., Rossi, C., (...), Antonelli, M. AISTech - Iron and Steel Technology Conference Proceedings, 2025 https://www.scopus.com/pages/publications/105008444563?origin=scopusAI

75. AI-Powered Digital Twins for Public Transportation: A Multi-agent Model for Transmilenio in Bogota Perez, M.-J., Chargui, T., Trentesaux, D. Studies in Computational Intelligence, 2025 https://www.scopus.com/pages/publications/105010625491?origin=scopusAI

76. Prototyping freight trains traffic management using multi-agent systems Cuppari, A., Guida, P.L., Martelli, M., (...), Zini, F. Proceedings - 1999 International Conference on Information Intelligence and Systems, ICIIS 1999, 1999 https://www.scopus.com/pages/publications/84902667874?origin=scopusAI

77. Smart Kavach: AI-Powered Railway Traffic Management System for Maximizing Section Throughput

Senthilnayaki, B., Niranjana Devi, P.C., Prethika, S. Proceedings - ICSES 2026: 5th International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems, 2026 https://www.scopus.com/pages/publications/105038355476?origin=scopusAI